

```
!pip install chromadb
!pip install sentence-transformers
!pip install langchain
```

Collecting chromadb

Downloading chromadb-1.5.6-cp39-abi3-manylinux_2_17_x86_64.manylinux2014_x86_64.whl.metadata (5.8 kB)

Requirement already satisfied: pydantic>=2.0 in /usr/local/lib/python3.12/dist-packages (5.8 kB)

Requirement already satisfied: pydantic-settings>=2.0 in /usr/local/lib/python3.12/dist-packages (5.8 kB)

Collecting pybase64>=1.4.1 (from chromadb)

Downloading pybase64-1.4.3-cp312-cp312-manylinux1_x86_64.manylinux2014_x86_64.whl (10 kB)

Requirement already satisfied: uvicorn>=0.18.3 in /usr/local/lib/python3.12/dist-packages (10 kB)

Requirement already satisfied: numpy>=1.22.5 in /usr/local/lib/python3.12/dist-packages (10 kB)

Requirement already satisfied: typing-extensions>=4.5.0 in /usr/local/lib/python3.12/dist-packages (10 kB)

Collecting onnxruntime>=1.14.1 (from chromadb)

Downloading onnxruntime-1.14.1-cp312-cp312-manylinux_2_17_x86_64.manylinux2014_x86_64.whl (11.5 MB)

Requirement already satisfied: opentelemetry-api>=1.2.0 in /usr/local/lib/python3.12/dist-packages (11.5 MB)

Collecting opentelemetry-exporter-otlp-proto-grpc>=1.2.0 (from chromadb)

Downloading opentelemetry-exporter-otlp-proto-grpc-1.20.0-py3-none-any.whl (10.5 kB)

Requirement already satisfied: opentelemetry-sdk>=1.2.0 in /usr/local/lib/python3.12/dist-packages (10.5 kB)

Requirement already satisfied: tokenizers>=0.13.2 in /usr/local/lib/python3.12/dist-packages (10.5 kB)

Collecting pypika>=0.48.9 (from chromadb)

Downloading pypika-0.51.1-py2.py3-none-any.whl.metadata (51 kB)

52.0/52.0 kB 1.0 MB/s eta 0:00:00

Requirement already satisfied: tqdm>=4.65.0 in /usr/local/lib/python3.12/dist-packages (52.0 kB)

Requirement already satisfied: overrides>=7.3.1 in /usr/local/lib/python3.12/dist-packages (52.0 kB)

Requirement already satisfied: importlib-resources in /usr/local/lib/python3.12/dist-packages (52.0 kB)

Requirement already satisfied: grpcio>=1.58.0 in /usr/local/lib/python3.12/dist-packages (52.0 kB)

Collecting bcrypt>=4.0.1 (from chromadb)

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Requirement already satisfied: typer>=0.9.0 in /usr/local/lib/python3.12/dist-packages (10 kB)

Collecting kubernetes>=28.1.0 (from chromadb)

Downloading kubernetes-30.0.0-py2.py3-none-any.whl.metadata (1.7 kB)

Requirement already satisfied: tenacity>=8.2.3 in /usr/local/lib/python3.12/dist-packages (1.7 kB)

Requirement already satisfied: pyyaml>=6.0.0 in /usr/local/lib/python3.12/dist-packages (1.7 kB)

Requirement already satisfied: mmh3>=4.0.1 in /usr/local/lib/python3.12/dist-packages (1.7 kB)

Requirement already satisfied: orjson>=3.9.12 in /usr/local/lib/python3.12/dist-packages (1.7 kB)

Requirement already satisfied: httpx>=0.27.0 in /usr/local/lib/python3.12/dist-packages (1.7 kB)

Requirement already satisfied: rich>=10.11.0 in /usr/local/lib/python3.12/dist-packages (1.7 kB)

Requirement already satisfied: jsonschema>=4.19.0 in /usr/local/lib/python3.12/dist-packages (1.7 kB)

Requirement already satisfied: packaging>=24.0 in /usr/local/lib/python3.12/dist-packages (1.7 kB)

Collecting pyproject_hooks (from build>=1.0.3->chromadb)

Downloading pyproject_hooks-1.2.0-py3-none-any.whl.metadata (1.3 kB)

Requirement already satisfied: anyio in /usr/local/lib/python3.12/dist-packages (1.3 kB)

Requirement already satisfied: certifi in /usr/local/lib/python3.12/dist-packages (1.3 kB)

Requirement already satisfied: httpcore==1.* in /usr/local/lib/python3.12/dist-packages (1.3 kB)

Requirement already satisfied: idna in /usr/local/lib/python3.12/dist-packages (1.3 kB)

Requirement already satisfied: h11>=0.16 in /usr/local/lib/python3.12/dist-packages (1.3 kB)

Requirement already satisfied: attrs>=22.2.0 in /usr/local/lib/python3.12/dist-packages (1.3 kB)

Requirement already satisfied: jsonschema-specifications>=2023.03.6 in /usr/local/lib/python3.12/dist-packages (1.3 kB)

Requirement already satisfied: referencing>=0.28.4 in /usr/local/lib/python3.12/dist-packages (1.3 kB)

Requirement already satisfied: rpds-py>=0.25.0 in /usr/local/lib/python3.12/dist-packages (1.3 kB)

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Requirement already satisfied: six>=1.9.0 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: python-dateutil>=2.5.3 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: websocket-client!=0.40.0,!=0.41.*,!=0.42.*,>=0.3.4 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: requests in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: requests-oauthlib in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: urllib3!=2.6.0,>=1.24.2 in /usr/local/lib/python3.12/dist-packages
Collecting durationpy>=0.7 (from kubernetes>=28.1.0->chromadb)
  Downloading durationpy-0.10.0-py3-none-any.whl metadata (3.40 bytes)
```

```
import chromadb
from sentence_transformers import SentenceTransformer
```

```
client = chromadb.Client()

collection = client.create_collection(name="rag_collection")
```

```
model = SentenceTransformer('all-MiniLM-L6-v2')
```

```
Loading weights: 100% 103/103 [00:00<00:00, 444.18it/s, Materializing param=pooler.dense.weight]
BertModel LOAD REPORT from: sentence-transformers/all-MiniLM-L6-v2
Key | Status | |
-----+-----+--
embeddings.position_ids | UNEXPECTED | |
```

Notes:

```
- UNEXPECTED :can be ignored when loading from different task/architecture; r
config.json: 100% 190/190 [00:00<00:00, 18.9kB/s]
```

```
documents = [
    "Artificial Intelligence is the simulation of human intelligence.",
    "Machine Learning is a subset of AI.",
    "Deep Learning uses neural networks.",
    "Natural Language Processing deals with text data.",
    "RAG combines retrieval and generation."
]

ids = ["doc1", "doc2", "doc3", "doc4", "doc5"]
```

```
embeddings = model.encode(documents)
```

```
collection.add(
    documents=documents,
    embeddings=embeddings,
    ids=ids
)
```

```
query = "What is AI?"

query_embedding = model.encode([query])

results = collection.query(
    query_embeddings=query_embedding,
    n_results=2
)

print(results)

{'ids': [['doc1', 'doc2']], 'embeddings': None, 'documents': [['Artificial Intel
```

```
for doc in results['documents'][0]:
    print(doc)
```

Artificial Intelligence is the simulation of human intelligence.
Machine Learning is a subset of AI.

```
def retrieve(query):
    query_embedding = model.encode([query])
    results = collection.query(query_embeddings=query_embedding, n_results=2)
    return results['documents'][0]

print(retrieve("Explain machine learning"))

['Machine Learning is a subset of AI.', 'Deep Learning uses neural networks.']
```